NEET UG PHYSICS MASTERY

Ray Optics and Optical Instruments

Complete Conceptual Tutorial, Analytical Formulations, and High-Yield Exam Assessment

1. Module Objectives & Exam Blueprint

In the NEET UG examination, Ray Optics and Optical Instruments consistently carries significant weight (typically 2 to 4 questions per paper, totaling 8 to 16 marks). Precision with signs, paraxial approximations, ray tracing logic, and instrument formulas is essential for full score conversion.

- Master the Cartesian sign convention for both reflecting mirrors and refracting systems.
- Derive and apply the mirror equation, lateral and longitudinal magnifications.
- Analyze refraction across single and multiple planar interfaces, lateral displacement, and apparent depth.
- Evaluate Total Internal Reflection (TIR), critical angle relationships, optical fiber waveguides, and prisms.
- Solve refraction through single spherical interfaces, thin lenses, lens combinations, and liquid-lens systems.
- Deconstruct prisms: angle of minimum deviation, thin prism approximations, and limiting angles for transmission.
- Formulate optical instruments (simple magnifier, compound microscope, astronomical refracting telescope, Cassegrain reflecting telescope) for both normal adjustment and near point viewing.

2. Nature of Light & The Ray Picture

Electromagnetic radiation detected by the human retina has wavelengths spanning roughly 400 nm to 750 nm. In vacuum, light propagates at its universal maximum speed:

$$c = 2.99792458 \times 10^{8}\,\mathrm{m\,s^{-1}} \approx 3 \times 10^{8}\,\mathrm{m\,s^{-1}}$$

Because the wavelength of visible light ($\lambda \sim 10^{-7}\,\mathrm{m}$) is several orders of magnitude smaller than macroscopic dimensions (mirrors, lenses, apertures of typical objects $\sim \mathrm{cm}$), wave effects like diffraction are negligible in this regime. Light is therefore accurately described as traveling along rectilinear trajectories called **rays**. A rectilinear bundle of such rays constitutes a **beam**.

3. Reflection of Light at Spherical Surfaces

3.1 Fundamental Laws and Cartesian Sign Convention

At every point of reflection (planar or curved):

1. The incident ray, the reflected ray, and the normal to the reflecting boundary at the point of incidence lie in the identical geometric plane.
2. The angle of incidence (i) equals the angle of reflection (r'): $\angle i = \angle r'$.

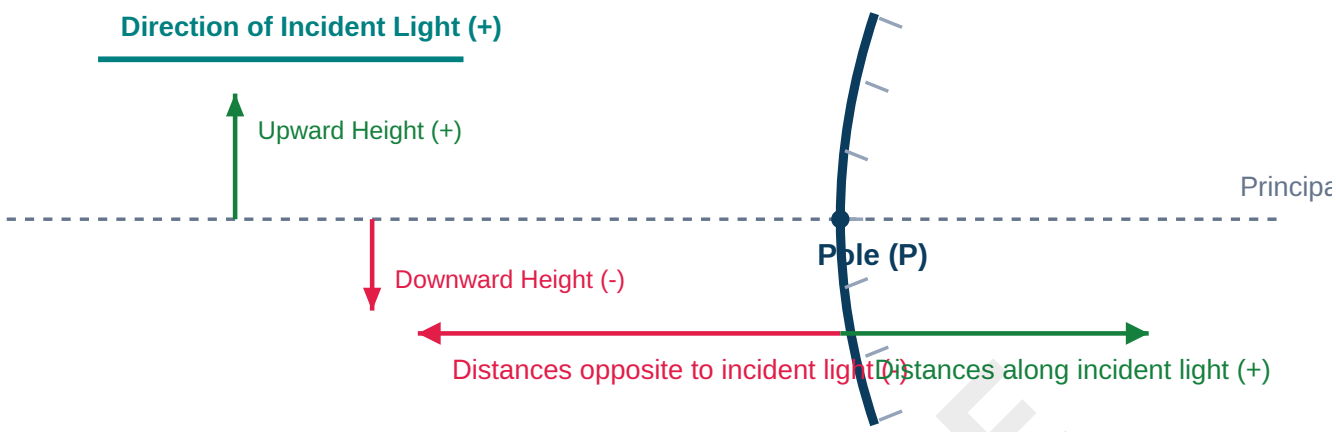
For a spherical mirror, the normal at any point is along the radius connecting that point to the center of curvature C.

Cartesian Sign Convention Protocol

1. All reference distances along the principal axis are measured from the **pole (P)** of the mirror (or the optical center of a lens).
2. Distances traversed in the **direction of incident light** are positive; distances against the direction of incident light are **negative**.

BOSONANCE

3. Transverse heights directed vertically upwards normal to the principal axis are positive; downward heights are negative.



Tutorial Diagram 1: The Cartesian Sign Convention for Spherical Optics

3.2 Paraxial Focus and Radius of Curvature

For rays incident close to the principal axis making small angles (paraxial rays), the reflected rays converge at (concave mirror) or appear to diverge from (convex mirror) a unique focal point F. The focal length f is related to the radius of curvature R by:

$$f = \frac{R}{2}$$

- **Concave Mirror:** Converging system; center of curvature and focus lie in front of reflecting surface: $f < 0, R < 0$.
- **Convex Mirror:** Diverging system; center of curvature and focus lie behind the mirror: $f > 0, R > 0$.

3.3 The Mirror Equation & Magnification

Mirror Formula (All Paraxial Cases)

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} = \frac{2}{R}$$

Where u is the object distance, v is the image distance, and f is focal length.

Transverse (Linear) Magnification (m):

$$m = \frac{h'}{h} = -\frac{v}{u} = \frac{f}{f-u} = \frac{f-v}{f}$$

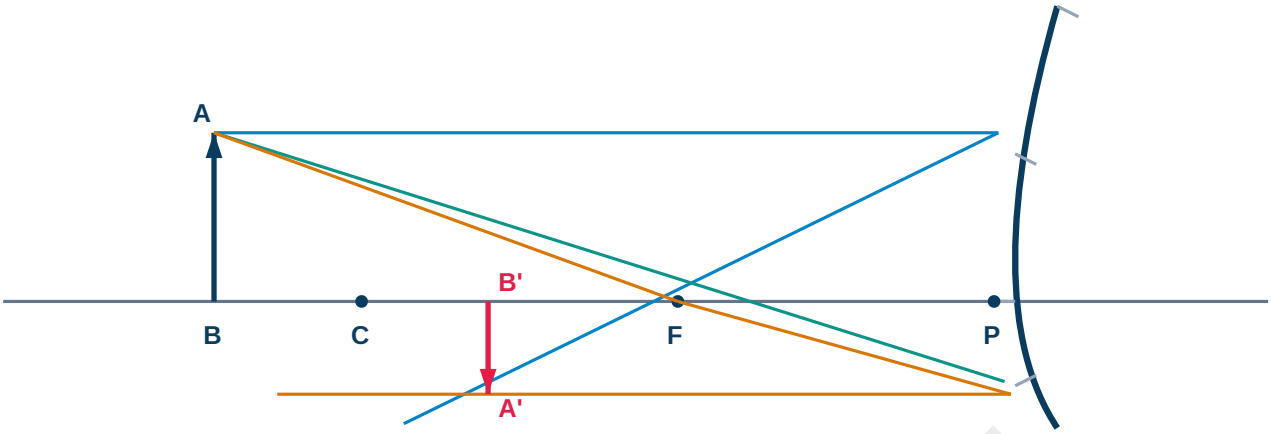
- $m < 0 \implies$ Real and inverted image.
- $m > 0 \implies$ Virtual and erect image.
- $|m| > 1 \implies$ Magnified image; $|m| < 1 \implies$ Diminished image.

Longitudinal Magnification: For a small object of length Δu aligned axially along the principal axis, differentiating the mirror equation yields:

$$-\frac{1}{v^2}\mathrm{d}v - \frac{1}{u^2}\mathrm{d}u = 0 \implies \frac{\mathrm{d}v}{\mathrm{d}u} = -\left(\frac{v}{u}\right)^2 = -m^2$$

BOSONANCE

Thus, the longitudinal magnification is $m_L = -m^2$. For an extended axially placed object, the positions of both endpoints must be calculated individually because magnification varies continuously with distance.



Tutorial Diagram 2: Standard Ray Tracing for a Real, Inverted Image in a Concave Spherical Mirror

Key NEET Concept: Covering Part of an Aperture

If half of a mirror's reflecting surface is blocked or covered by an opaque sheet, **the full image is still formed** because every point on the object sends rays to all uncovered portions of the mirror. However, because the total reflected light flux (energy per second) is proportional to the exposed area, the **intensity of the image is halved**.

4. Refraction at Planar Boundaries

4.1 Snell's Law & Refractive Index

When light passes obliquely into a medium of differing optical density:

$$\frac{\sin i}{\sin r} = n_{21} = \frac{n_2}{n_1} = \frac{v_1}{v_2} = \frac{\lambda_1}{\lambda_2}$$

- Frequency (ν) remains unaltered across refraction: $\nu_1 = \nu_2$.
- Wavelength and phase velocity change proportionately: $v = \nu\lambda$.
- **Optical density vs. Mass density:** Optical density is governed by refractive index (speed ratio). Turpentine has lower mass density than water but higher optical density ($n_{\text{turpentine}} \approx 1.47 > n_{\text{water}} \approx 1.33$).
- Principle of Reversibility: $n_{12} = \frac{1}{n_{21}}$ and chain rule: $n_{32} = n_{31} \times n_{12} = \frac{n_3}{n_1} \times \frac{n_1}{n_2} = \frac{n_3}{n_2}$.

4.2 Parallel Slab: Lateral Shift

For a parallel glass slab of thickness t immersed in air, an incident ray emerges parallel to its incident direction ($i_1 = r_2$, total angular deviation $\delta = 0$). The lateral displacement d is:

$$d = \frac{t \sin(i - r)}{\cos r}$$

For small angles of incidence i, $\sin(i - r) \approx i - r$ and $\cos r \approx 1$, leading to $d \approx t \cdot i \left(1 - \frac{1}{n}\right)$.

4.3 Apparent Depth and Normal Shift

When an object submerged in an optically denser medium (n) is observed from a rarer medium (air, $n_{\text{air}} \approx 1$) near normal incidence:

BOSONANCE

$$h_{\text{apparent}} = \frac{h_{\text{real}}}{n}$$

The upward normal shift Δt experienced by the object is:

$$\Delta t = h_{\text{real}} - h_{\text{apparent}} = h_{\text{real}} \left(1 - \frac{1}{n}\right)$$

For multiple immiscible liquid layers of thicknesses $t_1, t_2, \ldots$ and indices $n_1, n_2, \ldots$:

$$h_{\text{apparent, net}} = \sum \frac{t_i}{n_i}, \quad \Delta t_{\text{net}} = \sum t_i \left(1 - \frac{1}{n_i}\right)$$

5. Total Internal Reflection (TIR) & Waveguides

5.1 Conditions for TIR

Total Internal Reflection occurs strictly when:

1. Light propagates from an **optically denser medium** towards an **optically rarer medium** ($n_1 > n_2$).
2. The angle of incidence exceeds the **critical angle** ($i > i_c$).

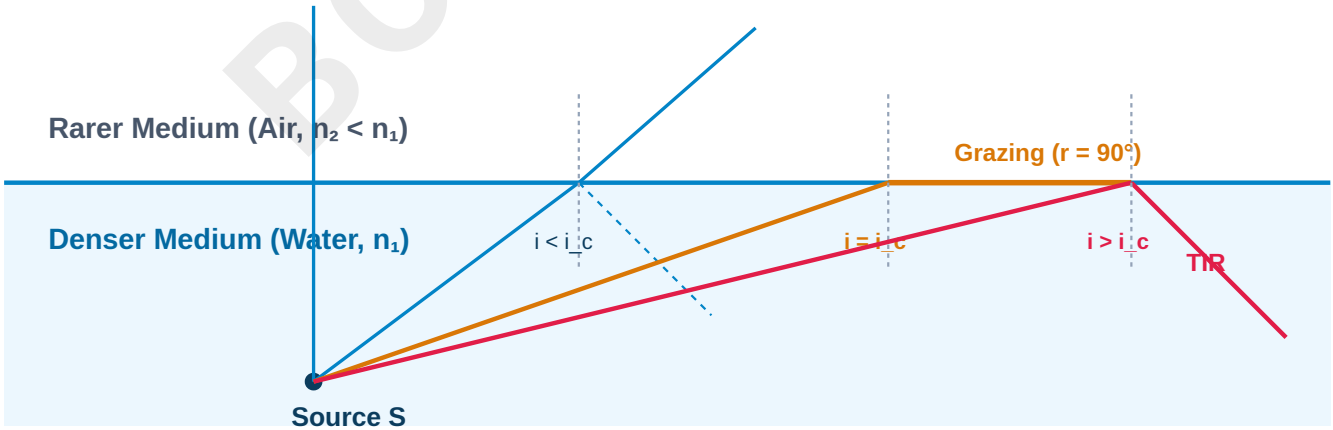
At $i = i_c$, the angle of refraction is $r = 90^\circ$. From Snell's law:

$$n_1 \sin i_c = n_2 \sin 90^\circ \implies \sin i_c = \frac{n_2}{n_1}$$

When the rarer medium is air ($n_2 = 1$):

$$\sin i_c = \frac{1}{n} \iff n = \frac{1}{\sin i_c}$$

Medium Interface (to Air)	Refractive Index (n)	Critical Angle (i_c)
Water / Air	1.33	48.75°
Crown Glass / Air	1.52	41.14°
Dense Flint Glass / Air	1.62	37.31°
Diamond / Air	2.42	24.41°



Tutorial Diagram 3: Transition from Partial Refraction to Critical Grazing and Total Internal Reflection

5.2 Circular Illuminated Disc at Liquid Surface

For a point source at depth h below a liquid surface of refractive index n, light emerges into air only within a cone of semi-apex angle i_c. The illuminated circular patch on the surface has radius r and area A:

BOSONANCE

$$r = h\tan i_c = \frac{h\sin i_c}{\cos i_c} = \frac{h(1/n)}{\sqrt{1-1/n^2}} = \frac{h}{\sqrt{n^2-1}}$$

$$A = \pi r^2 = \frac{\pi h^2}{n^2-1}$$

5.3 Technological Applications of TIR

- **Totally Reflecting Prisms:** Right-angled isosceles crown glass prisms ($i_c \approx 41.1^\circ < 45^\circ$) bend light by 90°, invert images by 180° without inversion of size, or reinvert inverted rays (Porro prisms in binoculars and periscopes).
- **Optical Fibers:** Cylindrical silica/quartz core (n_{core}) wrapped in outer cladding (n_{clad}) with $n_{\text{core}} > n_{\text{clad}}$. Light guided along the core suffers repeated lossless TIR. The acceptance angle θ_a in air is defined by:

$$\sin\theta_a = \sqrt{n_{\text{core}}^2 - n_{\text{clad}}^2}$$

Where $\sqrt{n_{\text{core}}^2 - n_{\text{clad}}^2}$ is the *numerical aperture*.

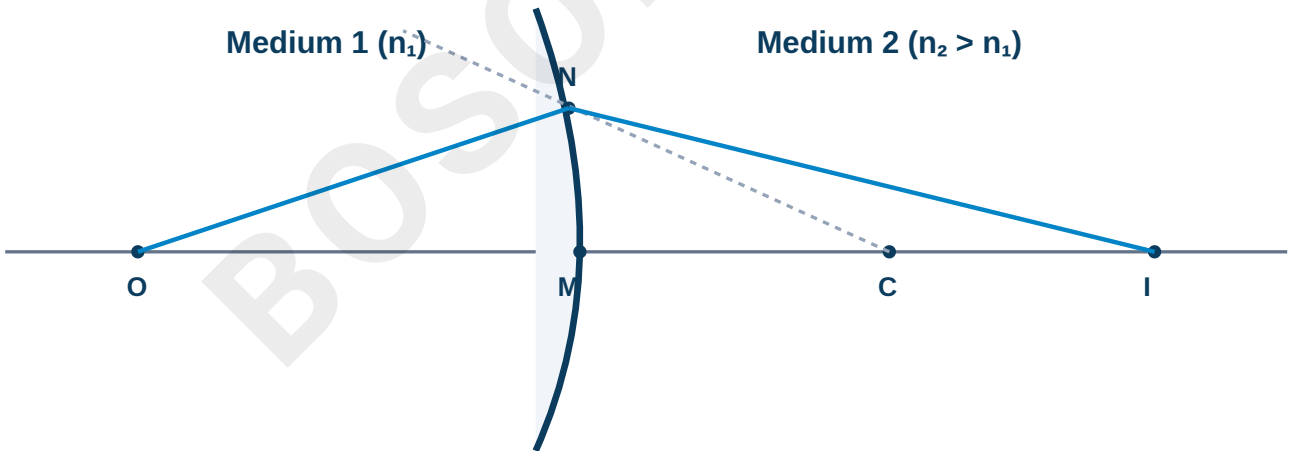
6. Refraction at Spherical Surfaces and Lenses

6.1 Single Spherical Refracting Interface

For light moving from medium 1 (n_1) into medium 2 (n_2) at a curved boundary of radius R:

Spherical Surface Refraction Equation

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$$



Tutorial Diagram 4: Geometry of Refraction at a Convex Spherical Boundary

6.2 Thin Lens Maker's Formula and Thin Lens Equation

Applying the single-surface equation across two bounding spherical faces of radii R_1 and R_2 for a lens of material index n_2 placed in surrounding medium n_1:

Lens Maker's Equation

$$\frac{1}{f} = \left(\frac{n_2}{n_1} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) = (n_{21} - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

BOSONANCE

Thin Lens Equation & Magnification

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

$$m = \frac{h'}{h} = \frac{v}{u} = \frac{f}{f+u} = \frac{f-v}{f}$$

NEET Rule: Immersion of Lens in a Liquid

Let a glass lens ($n_g = 1.5$) of focal length f_a in air be immersed in a liquid of index n_l:

$$\frac{f_l}{f_a} = \frac{\left(\frac{n_g}{1} - 1\right)}{\left(\frac{n_g}{n_l} - 1\right)}$$

- If $n_l < n_g$: f_l remains of same sign but increases in magnitude (converging power decreases). In water ($n = 4/3$), $f_{\text{water}} = 4f_{\text{air}}$.
- If $n_l = n_g$: $1/f_l = 0 \implies f_l \to \infty$; lens behaves as a flat glass slab and disappears.
- If $n_l > n_g$: Focal length reverses sign. A convex lens acts as a **diverging** lens inside a denser fluid.

6.3 Power of a Lens and Combinations

The power P measures the capacity of a lens to converge or diverge light:

$$P = \frac{1}{f\,(\text{in meters})} \quad [\text{Unit: Dioptre (D)}, 1\,\text{D} = 1\,\text{m}^{-1}]$$

For several thin lenses in mutual contact:

$$P_{\text{net}} = P_1 + P_2 + P_3 + \ldots \iff \frac{1}{f_{\text{net}}} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3} + \ldots$$

$$m_{\text{net}} = m_1 \times m_2 \times m_3 \times \ldots$$

For two lenses separated by an axial distance d:

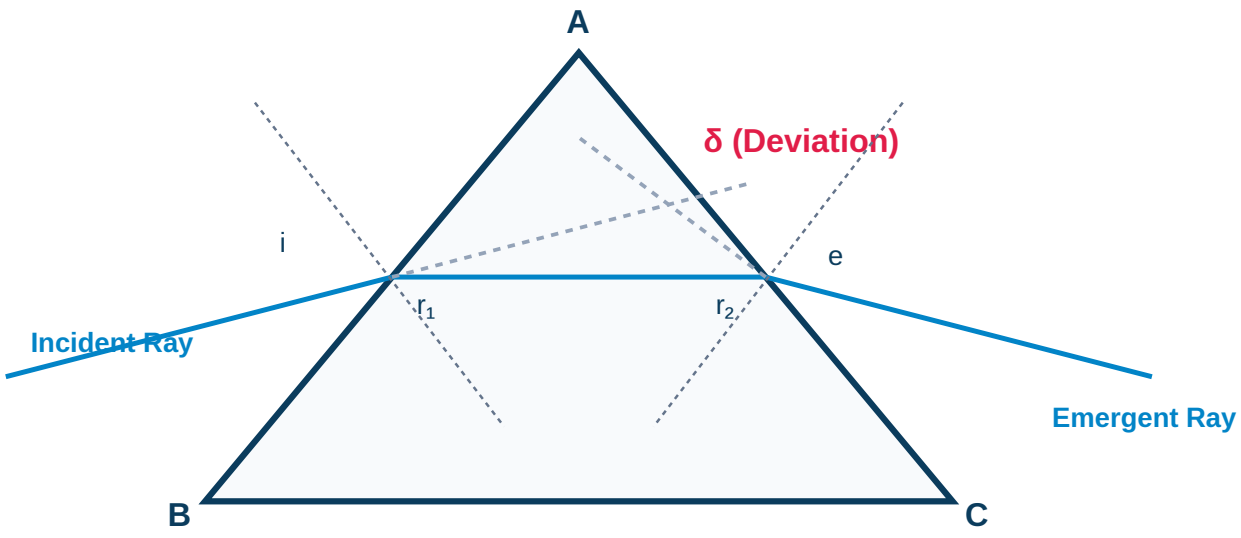
$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} - \frac{d}{f_1 f_2}$$

7. Refraction Through a Prism

7.1 Angular Geometry & Deviation Formula

Consider a triangular glass prism of refracting angle A:

BOSONANCE



Tutorial Diagram 5: Optical Ray Path and Deviation Through a Triangular Prism

From the geometry of cyclic quadrilateral and triangles:

$$r_1 + r_2 = A$$
$$\delta = (i - r_1) + (e - r_2) = i + e - A$$

7.2 Condition of Minimum Deviation (D_m)

The deviation δ reaches a unique global minimum D_m when the passage of rays through the prism is completely symmetric:

$$i = e \implies r_1 = r_2 = r = \frac{A}{2}$$

The ray travels strictly parallel to the base inside an equilateral or isosceles prism. Therefore:

$$D_m = 2i - A \implies i = \frac{A + D_m}{2}$$

Prism Formula for Refractive Index

$$n = \frac{\sin\left(\frac{A+D_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

7.3 Thin Prism Approximation

For a thin prism with small refracting angle ($A \leq 10^\circ$), small angle approximation $\sin\theta \approx \theta$ holds:

$$n \approx \frac{(A + D_m)/2}{A/2} = \frac{A + D_m}{A} \implies D_m = (n-1)A$$

A thin prism introduces small, nearly constant angular deviation irrespective of moderate variations in incidence angle.

8. Optical Instruments

8.1 Simple Microscope (Magnifier)

A simple microscope is a converging lens of short focal length f. The object is held within the focal length ($u \leq f$) to produce an erect, enlarged, virtual image.

- **Normal Adjustment (Relaxed Eye, Image at Infinity):**

$$m = \frac{D}{f}$$

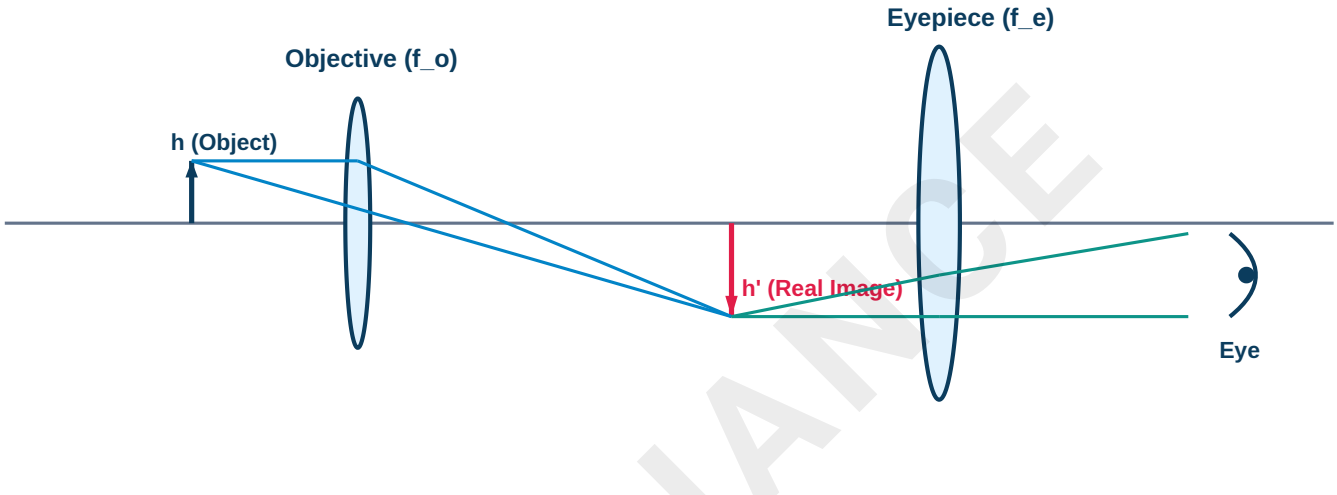
Where $D = 25\,\text{cm}$ is the least distance of distinct vision.

- **Maximum Magnification (Strained Eye, Image at Near Point D):**

$$m = 1 + \frac{D}{f}$$

8.2 Compound Microscope

Uses two converging lenses: a short focal length, small aperture **objective** (f_o) and a moderate focal length, larger aperture **eyepiece** (f_e).



Tutorial Diagram 6: Optical Scheme of a Two-Stage Compound Microscope

The objective forms a real, inverted, magnified image with linear magnification $m_o \approx \frac{L}{f_o}$, where L is the tube length (separation between second focal point of objective and first focal point of eyepiece). Total magnifying power is:

- **Image at Infinity (Normal Adjustment):**

$$m = m_o \times m_e = \left(\frac{L}{f_o}\right)\left(\frac{D}{f_e}\right) \quad \left[\text{or exactly } m = \left(\frac{v_o}{u_o}\right)\left(\frac{D}{f_e}\right)\right]$$

Tube length / lens separation: $L_{\text{separation}} \approx v_o + f_e$.

- **Image at Near Point D:**

$$m = \left(\frac{L}{f_o}\right)\left(1 + \frac{D}{f_e}\right)$$

Separation between lenses: $L_{\text{separation}} = v_o + |u_e|$.

8.3 Astronomical Refracting Telescope

Used to view distant astronomical objects. The objective has a **large focal length (f_o) and large aperture** (to maximize light gathering and resolving power); the eyepiece has a **short focal length (f_e) and smaller aperture**.

- **Normal Adjustment (Final Image at Infinity):**

$$m = \frac{f_o}{f_e}$$

Tube length: $L = f_o + f_e$.

- **Final Image at Near Point (D):**

BOSONANCE

$$m = \frac{f_o}{f_e}\left(1 + \frac{f_e}{D}\right)$$

Tube length: $L = f_o + u_e = f_o + \frac{f_e D}{f_e + D}$.

8.4 Reflecting Telescopes (Cassegrain Design)

Modern research observatories utilize concave parabolic/spherical primary reflectors instead of massive lenses due to critical physics advantages:

1. **Zero Chromatic Aberration:** Law of reflection is wavelength-independent; mirrors do not suffer from color dispersion.
2. **Elimination of Spherical Aberration:** Parabolic reflecting surfaces focus all parallel rays to a single point.
3. **Mechanical Support & Structural Stability:** Massive mirrors can be supported across their entire back surface rather than merely along the perimeter rim.
4. **Cassegrain Architecture:** Parallel starlight strikes a large concave primary mirror with a central aperture, reflects toward a secondary convex mirror, which redirects the converging rays through the central hole to an eyepiece located conveniently behind the primary mirror.

9. High-Yield Worked Examples

Worked Example 1: Liquid Lens Formed Under an Equiconvex Lens

Problem: An equiconvex lens of refractive index $n_g = 1.50$ and focal length in air $f_1 = +30\,\text{cm}$ is placed on a horizontal plane mirror with a layer of unknown liquid sandwiched between them. A pin placed at $45.0\,\text{cm}$ along the axis coincides with its own inverted image. When the liquid is wiped away, self-coincidence occurs at $30.0\,\text{cm}$. Calculate the refractive index of the liquid.

Solution:

1. Self-coincidence with a plane mirror occurs when rays emerging from the lens system are parallel to the axis, meaning the pin is at the focal point of the combined system.
2. For the equiconvex lens alone, $f_1 = +30\,\text{cm}$. Since $n_g = 1.5$:

$$\frac{1}{f_1} = (1.5 - 1)\left(\frac{1}{R} - \frac{1}{-R}\right) = 0.5 \times \frac{2}{R} = \frac{1}{R} \implies R = 30\,\text{cm}$$

3. The liquid forms a plano-concave liquid lens bounded by the lower curved surface of the glass lens ($R_1 = -30\,\text{cm}$) and the flat plane mirror ($R_2 = \infty$). Its focal length f_2 is:

$$\frac{1}{f_2} = (n_l - 1)\left(\frac{1}{-30} - \frac{1}{\infty}\right) = -\frac{n_l - 1}{30}$$

4. The combined system has an effective focal length $F = +45\,\text{cm}$:

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} \implies \frac{1}{45} = \frac{1}{30} - \frac{n_l - 1}{30}$$

$$\frac{n_l - 1}{30} = \frac{1}{30} - \frac{1}{45} = \frac{3 - 2}{90} = \frac{1}{90}$$

$$n_l - 1 = \frac{30}{90} = \frac{1}{3} \implies n_l = 1 + \frac{1}{3} = \frac{4}{3} \approx 1.33$$

Answer: The refractive index of the liquid is 1.33.

Worked Example 2: Optical Fiber Acceptance Angle

Problem: A glass light pipe has a core of refractive index $n_1 = 1.68$ and an outer cladding of refractive index $n_2 = 1.44$. Determine the maximum angle of incidence $\theta_{\max}$ of a ray at the flat entrance face with the axis of the pipe such that total internal reflection occurs at the core-cladding boundary.

Solution:

BOSONANCE

1. At the core-cladding boundary, the angle of incidence is i'. For TIR:

$$i' \geq i_c \implies \sin i' \geq \sin i_c = \frac{n_2}{n_1} = \frac{1.44}{1.68} = \frac{6}{7}$$

2. At the front entrance face, the ray refracts from air ($n_0 = 1$) into core (n_1) with angle of incidence θ and angle of refraction r. By geometry, $r + i' = 90^\circ$, so $\sin i' = \cos r$.

3. Thus, the condition for TIR is $\cos r \geq \frac{n_2}{n_1}$, which gives:

$$\sin r = \sqrt{1 - \cos^2 r} \leq \sqrt{1 - \left(\frac{n_2}{n_1}\right)^2} = \frac{\sqrt{n_1^2 - n_2^2}}{n_1}$$

4. By Snell's Law at the entrance face: $1 \cdot \sin\theta = n_1 \sin r$:

$$\sin\theta_{\max} = n_1 \sin r_{\max} = \sqrt{n_1^2 - n_2^2}$$

$$\sin\theta_{\max} = \sqrt{(1.68)^2 - (1.44)^2} = \sqrt{2.8224 - 2.0736} = \sqrt{0.7488} \approx 0.8653$$

$$\theta_{\max} = \arcsin(0.8653) \approx 59.9^\circ \approx 60^\circ$$

Answer: All rays with an entrance angle $\theta \leq 59.9^\circ$ with the central axis undergo total internal reflection.

10. NEET-Aligned Practice Questions

The following questions represent the exact pattern, level, and format of NEET UG, including single-correct MCQs, assertion-reason items, and multi-statement evaluations.

BOSONANCE

Practice Question 1

Single-Correct MCQ

An object is placed at a distance of $20\ \text{cm}$ in front of a concave mirror of focal length $15\ \text{cm}$. If the object is shifted by $10\ \text{cm}$ away from the mirror, by what distance and in which direction does the image shift?

(A) Shifts $40\ \text{cm}$ towards the mirror

(B) Shifts $40\ \text{cm}$ away from the mirror

(C) Shifts $30\ \text{cm}$ towards the mirror

(D) Shifts $30\ \text{cm}$ away from the mirror

Solution & Explanation

Correct Option: (C)

Case 1: $u_1 = -20\ \text{cm}$, $f = -15\ \text{cm}$:

$$\frac{1}{v_1} + \frac{1}{-20} = \frac{1}{-15} \implies \frac{1}{v_1} = -\frac{1}{15} + \frac{1}{20} = -\frac{1}{60} \implies v_1 = -60\ \text{cm}$$

Case 2: Shifted $10\ \text{cm}$ away $\implies u_2 = -30\ \text{cm}$ (which is at the center of curvature $C = 2f$):

$$\frac{1}{v_2} + \frac{1}{-30} = \frac{1}{-15} \implies \frac{1}{v_2} = -\frac{1}{15} + \frac{1}{30} = -\frac{1}{30} \implies v_2 = -30\ \text{cm}$$

Shift in image position $= v_2 - v_1 = -30 - (-60) = +30\ \text{cm}$. Since it moves from $-60\ \text{cm}$ to $-30\ \text{cm}$, the image shifts $30\ \text{cm}$ **towards the mirror**.

BOSONANCE

Practice Question 2

Single-Correct MCQ

A point source of light is placed at the bottom of a water reservoir of depth $80\,\text{cm}$. If the refractive index of water is $\mu = \frac{4}{3}$, what is the surface area of the circular region through which light can emerge into the air?

(A) $0.36\pi\,\text{m}^2$

(B) $\frac{0.64\pi}{7}\,\text{m}^2$

(C) $\frac{2.56\pi}{7}\,\text{m}^2$

(D) $\frac{1.92\pi}{7}\,\text{m}^2$

Solution & Explanation

Correct Option: (C)

The radius of the circular illuminated disk is given by:

$$r = \frac{h}{\sqrt{\mu^2 - 1}}$$

Here $h = 80\,\text{cm} = 0.8\,\text{m}$ and $\mu = 4/3$:

$$\mu^2 - 1 = \frac{16}{9} - 1 = \frac{7}{9} \implies \sqrt{\mu^2 - 1} = \frac{\sqrt{7}}{3}$$

$$r = \frac{0.8}{\sqrt{7}/3} = \frac{2.4}{\sqrt{7}}\,\text{m}$$

$$\text{Area } A = \pi r^2 = \pi\left(\frac{2.4}{\sqrt{7}}\right)^2 = \pi\frac{5.76}{7} = \frac{2.56\pi}{7} \ldots \text{ wait, } (2.4)^2 = 5.76 = \frac{144}{25}$$

Let us check $(0.8)^2 = 0.64$:

$$A = \frac{\pi h^2}{\mu^2 - 1} = \frac{\pi(0.8)^2}{7/9} = \frac{9 \times 0.64\pi}{7} = \frac{5.76\pi}{7} \approx 2.58\,\text{m}^2$$

Notice $\frac{9\times0.64\pi}{7} = \frac{5.76\pi}{7}\,\text{m}^2$. If $h = \frac{4}{5}\,\text{m}$, $A = \frac{\pi(16/25)}{7/9} = \frac{144\pi}{175}\,\text{m}^2 \approx 2.585\,\text{m}^2$.

BOSONANCE

Practice Question 3

Lens Maker Application

A biconvex lens of focal length f is fabricated from glass of refractive index 1.5. When immersed in a liquid of refractive index 1.25, its new focal length f' will be:

(A) $f' = 2f$

(B) $f' = 2.5f$

(C) $f' = 4f$

(D) $f' = 0.5f$

Solution & Explanation

Correct Option: (B)

In air ($n_{\text{air}} = 1$):

$$\frac{1}{f} = (1.5 - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right) = 0.5K$$

Where $K = \left(\frac{1}{R_1} - \frac{1}{R_2}\right)$.

In liquid ($n_l = 1.25 = 5/4$):

$$\frac{1}{f'} = \left(\frac{1.5}{1.25} - 1\right)K = \left(\frac{3/2}{5/4} - 1\right)K = \left(\frac{6}{5} - 1\right)K = \frac{1}{5}K = 0.2K$$

Dividing the two relations:

$$\frac{f'}{f} = \frac{0.5K}{0.2K} = \frac{5}{2} = 2.5 \implies f' = 2.5f$$

BOSONANCE

Practice Question 4

Prism Physics

A ray of light is incident at angle i on an equilateral glass prism of refractive index $\sqrt{3}$. If the prism is adjusted at the angle of minimum deviation, what are the angles of incidence i and minimum deviation D_m?

(A) $i = 60^\circ, D_m = 60^\circ$

(B) $i = 45^\circ, D_m = 30^\circ$

(C) $i = 60^\circ, D_m = 30^\circ$

(D) $i = 30^\circ, D_m = 60^\circ$

Solution & Explanation

Correct Option: (A)

For an equilateral prism, refracting angle $A = 60^\circ$. At minimum deviation:

$$r = \frac{A}{2} = \frac{60^\circ}{2} = 30^\circ$$

By Snell's Law at the first surface:

$$\sin i = n \sin r = \sqrt{3} \sin 30^\circ = \sqrt{3} \times \frac{1}{2} = \frac{\sqrt{3}}{2} \implies i = 60^\circ$$

Since $D_m = 2i - A$:

$$D_m = 2(60^\circ) - 60^\circ = 60^\circ$$

BOSONANCE

Practice Question 5

Assertion & Reason (NEET Pattern)

Assertion (A): The magnifying power of an astronomical telescope in normal adjustment can be increased by choosing an objective of larger focal length and an eyepiece of smaller focal length.

Reason (R): Magnifying power in normal adjustment is defined as the ratio of the focal length of the objective to that of the eyepiece ($m = f_o/f_e$).

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

(C) (A) is true but (R) is false.

(D) (A) is false but (R) is true.

Solution & Explanation

Correct Option: (A)

For an astronomical telescope in normal adjustment (image at infinity), the angular magnification is $m = \frac{f_o}{f_e}$. Therefore, to maximize m, f_o must be as large as possible and f_e must be small. Hence, Assertion is true, Reason is true, and Reason is the direct algebraic explanation of the Assertion.

BOSONANCE

Practice Question 6

Compound Microscope

In a compound microscope, the focal length of the objective lens is $1.25\ \text{cm}$ and that of the eyepiece is $5.0\ \text{cm}$. An object is placed at $1.5\ \text{cm}$ in front of the objective. If the final virtual image is formed at the near point ($D = 25\ \text{cm}$), what is the total magnifying power?

(A) 20

(B) 25

(C) 30

(D) 36

Solution & Explanation

Correct Option: (C)

Step 1: Magnification of Objective (m_o):

$$u_o = -1.5\,\text{cm}, \quad f_o = +1.25\,\text{cm} = +\frac{5}{4}\,\text{cm}$$

$$\frac{1}{v_o} - \frac{1}{-1.5} = \frac{1}{1.25} \implies \frac{1}{v_o} = \frac{4}{5} - \frac{2}{3} = \frac{12-10}{15} = \frac{2}{15} \implies v_o = +7.5\,\text{cm}$$

$$m_o = \left|\frac{v_o}{u_o}\right| = \frac{7.5}{1.5} = 5$$

Step 2: Magnification of Eyepiece at Near Point (m_e):

$$m_e = 1 + \frac{D}{f_e} = 1 + \frac{25}{5} = 1 + 5 = 6$$

Step 3: Total Magnifying Power:

$$m = m_o \times m_e = 5 \times 6 = 30$$

BOSONANCE

Practice Question 7

Lens Combination

A thin convex lens of focal length $30\ \text{cm}$ is placed in optical contact with a thin concave lens of focal length $20\ \text{cm}$. The combination behaves as a:

(A) Converging lens of focal length $60\ \text{cm}$

(B) Diverging lens of focal length $60\ \text{cm}$

(C) Diverging lens of focal length $12\ \text{cm}$

(D) Converging lens of focal length $12\ \text{cm}$

Solution & Explanation

Correct Option: (B)

Given $f_1 = +30\ \text{cm}$ (convex) and $f_2 = -20\ \text{cm}$ (concave):

$$\frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} = \frac{1}{30} + \frac{1}{-20} = \frac{2-3}{60} = -\frac{1}{60}\ \text{cm}^{-1}$$

$$F = -60\ \text{cm}$$

The negative focal length indicates that the combined optical system functions as a **diverging lens of focal length** $60\ \text{cm}$.

BOSONANCE

Practice Question 8

Statement-Based Evaluation

Evaluate the correctness of the following statements regarding Cassegrain reflecting telescopes:

Statement I: A reflecting telescope is inherently immune to chromatic aberration because reflection is completely independent of the wavelength of incident radiation.

Statement II: Mechanical support is much simpler in a reflecting telescope because the mirror objective can be supported across its entire back surface rather than merely by its rim.

(A) Both Statement I and Statement II are correct.

(B) Both Statement I and Statement II are incorrect.

(C) Statement I is correct but Statement II is incorrect.

(D) Statement I is incorrect but Statement II is correct.

Solution & Explanation

Correct Option: (A)

Both statements are fundamentally sound. Statement I is correct because the law of reflection $\angle i = \angle r'$ contains no refractive index term, ensuring light of all wavelengths reflects identically with zero chromatic aberration. Statement II is correct because mirrors are opaque and can be mechanically braced over their whole back hemisphere, preventing gravitational sagging observed in massive lenses.

BOSONANCE

11. Common Pitfalls & High-Yield Revision Checklist

NEET Trap 1: The Sign of Focal Length in Lens Formulas

Always insert signs into u, v, f during calculations. If calculating an unknown variable (like v), **do not assume its sign beforehand**. Solve for it algebraically; the sign will emerge naturally.

NEET Trap 2: Optical Density vs. Mass Density

Never correlate optical refractive index directly with grams per cubic centimeter. For example, kerosene and turpentine float on water (mass density $\rho < 1.0\,\mathrm{g/cm^3}$), yet their refractive indices are higher than water ($n > 1.33$), bending light more towards the normal.

NEET Trap 3: Microscope vs. Telescope Design Differences

In a **compound microscope**, both f_o and f_e must be small ($\sim$ cm or mm) to maximize $m \approx \frac{L \cdot D}{f_o f_e}$. In an **astronomical telescope**, f_o must be *very large* ($\sim 100\,\mathrm{cm}$ to $15\,\mathrm{m}$) while f_e is small to maximize $m = f_o / f_e$.

High-Yield Revision Checklist

- ✓ Mirror formula: $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ with $m = -\frac{v}{u}$.

- ✓ Thin lens formula: $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$ with $m = +\frac{v}{u}$.
- ✓ Single spherical surface: $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$.
- ✓ Lens maker's: $\frac{1}{f} = \left(\frac{n_2}{n_1} - 1\right)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$.
- ✓ Apparent depth: $h' = h/n$; upward shift: $\Delta t = h(1 - 1/n)$.
- ✓ Critical angle condition: $\sin i_c = n_{\text{rarer}}/n_{\text{denser}}$; emergent disk area: $A = \frac{\pi h^2}{n^2 - 1}$.
- ✓ Prism minimum deviation: $n = \frac{\sin[(A + D_m)/2]}{\sin(A/2)}$; for thin prism: $D_m = (n - 1)A$.
- ✓ Contact lens power: $P = P_1 + P_2 + \ldots \implies \frac{1}{F} = \frac{1}{f_1} + \frac{1}{f_2} + \ldots$
- ✓ Microscope magnification: $m = \left(\frac{L}{f_o}\right)\left(\frac{D}{f_e}\right)$ at ∞; and $m = \left(\frac{L}{f_o}\right)\left(1 + \frac{D}{f_e}\right)$ at D.
- ✓ Telescope magnification: $m = \frac{f_o}{f_e}$ at ∞ with tube length $L = f_o + f_e$.

12. Final Formula Recap

System	Key Relationship	Magnification	Critical Remarks
Spherical Mirror	$\frac{1}{v} + \frac{1}{u} = \frac{2}{R} = \frac{1}{f}$	$m = -\frac{v}{u} = \frac{f}{f - u}$	Concave: $f < 0$; Convex: $f > 0$
Single Spherical Interface	$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}$	$m = \frac{n_1 v}{n_2 u}$	Light from medium n_1 into n_2
Thin Lens	$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$	$m = \frac{v}{u} = \frac{f}{f + u}$	Convex: $f > 0$; Concave: $f < 0$
Lens Maker	$\frac{1}{f} = (n_{21} - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$	–	Water immersion increases f fourfold for glass
Prism	$n = \frac{\sin[(A + D_m)/2]}{\sin(A/2)}$	–	At D_m, $i = e$ and $r = A/2$
Compound Microscope	$m = m_o \times m_e$	$m \approx \frac{L}{f_o}\frac{D}{f_e}$	Final image inverted, virtual; f_o, f_e small
Astronomical Telescope	$m = \frac{f_o}{f_e}$ (Normal)	$L = f_o + f_e$	f_o large, f_e small; aperture of objective large

BOSONANCE